



Community detection via graph regularized symmetric nonnegative matrix tri-factorization

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Abstract

Identifying meaningful communities in complex networks is essential for uncovering functional structures in social, biological, and technological systems. Despite the success of Nonnegative Matrix Factorization (NMF) methods, most existing approaches fail to simultaneously preserve local structural patterns, enforce clear community separation, and effectively handle the sparsity of real-world graphs. To address these challenges, we propose *SPD-SNMTF*, a Structural Proximity and Diagonalized Symmetric Nonnegative Matrix Tri-Factorization framework that integrates three key components: (1) a hybrid similarity matrix combining explicit links and latent neighborhood affinities; (2) diagonal dominance constraints to enhance inter-community separation; and (3) graph Laplacian regularization to preserve topological smoothness. This unified approach jointly reconstructs the adjacency matrix while enforcing geometric and sparsity-aware constraints, resulting in more interpretable community detection with stable performance across a wide range of parameter settings. Experiments on eleven real-world datasets demonstrate that SPD-SNMTF consistently outperforms twelve state-of-the-art methods in terms of NMI, ACC, and ARI. Implementation details are available at <https://github.com/sohrabi94/SPD-SNMTF>.

Keywords Community detection · Symmetric nonnegative matrix tri-factorization · Diagonal dominance · Structural proximity · Interpretable clustering

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1 Introduction

Community detection means finding groups of closely connected nodes, called communities, within complex networks. It is a key technique in network analysis, with applications in social networks, biological systems, recommendation engines, and many other areas [1]. In these networks, communities are groups where the nodes are densely linked to each other but only loosely connected to the rest of the network. For example, in social networks, communities often represent friend circles or shared-interest groups [2], while in biological networks, they can correspond to protein complexes or metabolic pathways [3]. Detecting these hidden structures provides important insights, such as making better product recommendations in e-commerce [4], identifying functional modules in genomics [5], or improving routing in communication networks [6]. As networks become larger and more complex, there is an urgent need for community detection methods that are both accurate and efficient [7].

Many approaches have been proposed for community detection, including spectral clustering, modularity optimization, probabilistic models, and label propagation. Among them, NMF has gained considerable attention due to its interpretability, nonnegativity constraints, and flexibility in incorporating prior knowledge such as node attributes and network topology. By factorizing a nonnegative matrix derived from the network structure into low-dimensional factors, NMF provides an effective framework for uncovering latent community memberships [8, 9]. Recent extensions, including asymmetric and graph regularized NMF variants, further enhance clustering performance by preserving directional relationships and local topological structures [1, 10]. In its standard form, NMF approximates an input matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ as $\mathbf{M} \approx \mathbf{X}\mathbf{Y}$, where $\mathbf{X}$ encodes node-community memberships and $\mathbf{Y}$ captures the contribution of each community to the network reconstruction.

While standard NMF is effective for many data types, it does not explicitly preserve the symmetry of adjacency matrices in undirected networks, which can lead to less accurate and less interpretable clustering results. To address this limitation, *Symmetric NMF (SymNMF)* is specifically designed for undirected graphs by factorizing a symmetric adjacency matrix $\mathbf{M}$ as $\mathbf{X}\mathbf{X}^T$, thereby preserving nonnegativity and symmetry [11]. This formulation naturally captures reciprocal node relationships and improves clustering consistency. Recent extensions, including weighted and graph regularized SymNMF variants, further enhance performance by incorporating node attributes and preserving global and local structural information [8, 12].

However, SymNMF can still be limited in its expressiveness because it uses only a single factor matrix $\mathbf{X}$ and cannot explicitly model the relationships between communities. In many real-world networks, nodes may belong to multiple overlapping communities or exhibit varying degrees of interaction strength across groups. To address these challenges, *Nonnegative Matrix Tri-Factorization (NMTF)* introduces a third factor matrix $\mathbf{Y}$, enabling finer-grained community detection through the decomposition $\mathbf{M} \approx \mathbf{X}\mathbf{Y}\mathbf{Z}$. This richer formulation allows $\mathbf{Y}$ to capture inter-community correlations and interaction patterns, improving the ability to detect overlapping or hierarchical community structures. As a result,

NMTF has been shown to produce more flexible and accurate community assignments, especially in complex networks where simple symmetric factorization is insufficient. Building on the advantages of NMTF, Symmetric NMTF (SNMTF) adapts the tri-factorization framework to undirected networks by reducing the number of unknown matrices and enforcing symmetry constraints. This approach enables efficient and meaningful analysis of community structures by maintaining the inherent symmetric relationships in the data [13]. One of the key advantages of using SNMTF is its ability to detect communities in adjacency matrices with specific structures that cannot be accurately factorized by SymNMF.

Although SNMTF improves upon NMF and SymNMF by preserving symmetry and reducing model complexity, several key challenges remain. Existing SNMTF methods often fail to preserve local structural proximity, leading to unstable or fragmented community assignments, particularly in sparse networks [10, 14]. Moreover, the lack of explicit mechanisms for enforcing community separation results in blurred boundaries and reduced interpretability [15]. In addition, most approaches do not fully exploit hybrid similarity measures that combine explicit links with latent affinities, limiting their ability to handle sparsity effectively [1, 16]. Finally, graph Laplacian regularization is not adequately integrated within the SNMTF framework, despite its proven effectiveness in preserving topological smoothness [10, 17]. These limitations motivate the following research questions: *Q1* How can SNMTF preserve local structural proximity while effectively handling sparsity? *Q2* How can clearer community separation be enforced to improve interpretability and robustness? *Q3* How can graph regularization be effectively integrated into symmetric tri-factorization?

To address the above challenges, we propose *SPD-SNMTF*, a *Structural Proximity and Diagonalized Symmetric Nonnegative Matrix Tri-Factorization* framework for community detection in sparse undirected networks. The proposed method makes three key contributions. First, we introduce a symmetric tri-factorization model that explicitly captures inter-community correlations through a dedicated community interaction matrix, while preserving the inherent symmetry of the network. Second, we design a hybrid similarity matrix that integrates explicit graph connections with latent neighborhood affinities, enabling effective preservation of local structural proximity and robust performance under high sparsity. Third, we incorporate a diagonal dominance constraint on the community interaction matrix to enforce clear community separation and enhance interpretability, together with graph Laplacian regularization to ensure topological smoothness. By unifying these components within a single optimization framework, SPD-SNMTF promotes a block-diagonal community structure, preserves fine-grained local patterns, and delivers robust and interpretable community detection results.

The remainder of this paper is organized as follows: Sect. 2 reviews related work in community detection, NMF, and graph regularization. Section 3 presents the proposed methodology, including the mathematical formulation and optimization algorithm. Section 4 describes the experimental setup and results. Finally, Sect. 5 concludes the paper and discusses future research directions.

2 Background and related works

2.1 Notation

The notation $\mathbb{R}_+^{m \times n}$ denotes the space of $m \times n$ nonnegative matrices. We represent a graph $G = (V, E)$ with node set $V = \{v_1, \dots, v_n\}$ and edge set $E \subseteq V \times V$ through its adjacency matrix $\mathbf{M} \in \mathbb{R}_+^{n \times n}$, where the entry M_{ij} denotes the connection strength between nodes v_i and v_j ($M_{ij} = 1$ for unweighted edges or $M_{ij} = w_{ij}$ for weighted graphs). The similarity matrix $\mathbf{S} \in \mathbb{R}_+^{n \times n}$ quantifies pairwise node similarities based on neighborhood structures or nodal attributes. These matrices collectively capture both the explicit connectivity ($\mathbf{M}$) and the implicit relational patterns essential for community analysis. The NMF framework factorizes $\mathbf{M} \approx \mathbf{X}\mathbf{Y}$ into two nonnegative matrices: the community membership matrix $\mathbf{X} \in \mathbb{R}_+^{n \times k}$, where X_{ip} quantifies node i 's membership in community p , and the community feature matrix $\mathbf{Y} \in \mathbb{R}_+^{k \times n}$, where Y_{pj} represents community p 's influence on node j . Key mathematical notations include the Frobenius norm $\|\cdot\|_F$, used to measure reconstruction error such as $\|\mathbf{M} - \mathbf{X}\mathbf{Y}\|_F^2$; the Hadamard product $\odot$, denoting element-wise multiplication; the $n \times n$ matrix of ones $\mathbf{1}_n$; the identity matrix $\mathbf{I}_n$; and the trace operator $\text{Tr}(\cdot)$, which returns the sum of the diagonal elements.

2.2 Related works

In the field of community detection, NMF and its advanced variants have been extensively utilized for network analysis, community detection, and graph clustering. These methods have evolved to address various challenges including preserving network symmetry, capturing local structures, and integrating additional information. The foundational NMF approach [18] factorizes a nonnegative input matrix $\mathbf{M} \in \mathbb{R}_+^{m \times n}$ into two nonnegative matrices $\mathbf{X} \in \mathbb{R}_+^{m \times k}$ (basis features) and $\mathbf{Y} \in \mathbb{R}_+^{k \times n}$ (coefficients) through the approximation $\mathbf{M} \approx \mathbf{X}\mathbf{Y}$, where $k \ll \min(m, n)$. Pioneering work by Wang et al. [19] adapted NMF for community detection using diffusion kernel similarity in biological networks, sparking subsequent algorithmic developments.

For undirected graphs with symmetric adjacency matrices $\mathbf{M} \in \mathbb{R}_+^{n \times n}$, Symmetric NMF (SymNMF) [11] enforces $\mathbf{M} \approx \mathbf{X}\mathbf{X}^\top$ where $\mathbf{X} \in \mathbb{R}_+^{n \times k}$ represents node-community memberships. This preserves matrix symmetry through the $\mathbf{X}\mathbf{X}^\top$ structure, aligning with undirected network reciprocity. Lou et al. [20] enhanced SymNMF's stability through optimized scaling factors in multiplicative updates. When SymNMF proves inadequate for complex community structures, NMTF [21] offers enhanced flexibility via three-matrix decomposition $\mathbf{M} \approx \mathbf{X}\mathbf{Y}\mathbf{Z}$ with $\mathbf{X} \in \mathbb{R}_+^{n \times k}$, $\mathbf{Y} \in \mathbb{R}_+^{k \times k}$, and $\mathbf{Z} \in \mathbb{R}_+^{k \times n}$. Wang et al. [13] developed Symmetric NMTF (SNMTF) to preserve undirected graph symmetry through the factorization $\mathbf{M} \approx \mathbf{X}\mathbf{Y}\mathbf{X}^\top$, where $\mathbf{Y}$ serves as the inter-community correlation matrix, with Y_{ij} quantifying interaction intensity between communities i and j .

It is worth noting that the community detection problem addressed in this work can also be viewed from the perspective of subspace clustering. Methods such as Structure-Constrained Low-Rank Representation (SCLRR) [22] aim to uncover low-dimensional subspace structures while enforcing structural constraints, which is conceptually related to our SPD-SNMTF framework. In particular, SPD-SNMTF seeks latent low-dimensional representations of nodes via tri-factorization while preserving local and global structural patterns, analogous to the way subspace clustering captures intrinsic data structures.

Graph Regularized NMF (GNMF) extends traditional NMF by incorporating manifold learning through a graph Laplacian term $\text{Tr}(\mathbf{X}^\top \mathbf{L} \mathbf{X})$ where $\mathbf{L} = \mathbf{D} - \mathbf{S}$. This preserves local geometric structure by enforcing connected nodes (with $S_{ij} > 0$) to have similar latent representations. Recent advances integrate local structure preservation through similarity graph construction. Tang et al. [17] unified global/local information in NMF models, while Berahmand et al. [23] developed adaptive local structure preservation. Pei et al. [16] applied these principles to NMTF, and Hajiveisheh et al. [1] proposed deep asymmetric NMF (DASNMF) for enhanced local structure retention. For attributed graphs, Arina et al. [24] combined NMTF with attribute-aware similarity metrics. Building on these, Yan et al. [25] introduced HNG, a higher-order proximity (HOP)-enhanced nonlinear NMF framework that integrates graph neural networks (GNNs) to address NMF's linearity limitations. Addressing the topology–content imbalance, Luo et al. [26] proposed two novel NMF variants: Double Encoder–Decoder NMF (DEDNMF) and Double Decoder NMF (DDNMF). DEDNMF employs dual encoder–decoders to adaptively fuse topology and node content, while DDNMF uses sparsity-constrained dual decoders for balanced reconstruction. To refine sparse regularization, Su et al. [27] introduced node centrality-weighted NMF (CWNMF), where the Gini impurity measures community membership sparsity for nodes and scale model with node centrality, enforcing influential nodes to exhibit clearer community assignments.

Similarity graph construction remains critical, with Jiyang et al. [28] employing dual hyper-graphs for higher-order dependencies, Sohrabi et al. [29] proposing global thresholding, and Chang et al. [30] fusing multiple graphs via fusion terms. However, these methods struggle with sparse graph preservation, highlighting the need for approaches that simultaneously maintain sparsity patterns and community structures. Table 1 provides a comprehensive overview of related methods, considering key attributes in the NMF process.

3 Proposed method

3.1 Motivation

The growing availability of complex network data, where connections between nodes are often sparse and communities are not clearly separated, has revealed important limitations in traditional matrix factorization methods. Although techniques like SymNMF and SNMTF can produce interpretable results by breaking

Table 1 Comparison of related works

Method	Symmetry	Local	Tri-factor	Graph Reg.	Correlation	Hybrid	Sparsity
NMF	✗	✗	✗	✗	✗	✗	✗
SymNMF	✓	✗	✗	✗	✗	✗	✗
NMTF	✗	✗	✓	✗	✗	✗	✗
SNMTF	✓	✗	✓	✗	✗	✗	✗
GNMF	✗	✓	✗	✓	✗	✗	✗
DASNMF	✗	✓	✓	✓	✗	✓	✗
DEDNMF	✗	✓	✗	✓	✗	✗	✓
DDNMF	✗	✓	✗	✓	✗	✗	✓
HNG	✗	✓	✗	✓	✗	✓	✗
CWNMF	✗	✓	✗	✓	✗	✓	✓
Proposed method	✓	✓	✓	✓	✓	✓	✓

✓ indicates support; ✗ indicates lack of support. *Correlation* refers to community correlation, *Symmetry* to symmetry preservation, *Hybrid* to graph similarity construction using structural proximities, *Tri-Factor* to three-matrix factorization, and *Graph Reg.* to graph regularization

the network into simpler parts, they still struggle with several challenges. First, because the correlation matrices in these models are not restricted, they often cannot clearly distinguish strong connections within a community from weaker links between communities. This makes it hard to identify well-defined groups, especially in sparse networks [31]. Second, these methods have difficulty preserving local structural patterns, such as nodes that are similar because they share many neighbors, when the adjacency matrix does not show these connections directly [17]. Finally, they are sensitive to noise and can be easily affected by weakly connected or irrelevant parts of the network, which reduces the stability and accuracy of the detected communities [15].

To overcome these challenges, we propose *SPD-SNMTF*, a *Structural Proximity and Diagonalized Symmetric Nonnegative Matrix Tri-Factorization* approach that integrates three complementary components within a unified model. First, our method decomposes the adjacency matrix into symmetric factor matrices, enabling the extraction of inter-community relationships while preserving the undirected nature of the graph. Second, it constructs a hybrid similarity matrix that blends explicit edges with cosine-based neighborhood affinities, improving the detection of local structural patterns even in sparse networks. Third, a diagonal dominance constraint is applied to the community interaction matrix to ensure that intra-community correlations remain stronger than inter-community links, which sharpens community boundaries. Additionally, graph Laplacian regularization is incorporated to encourage smooth node embeddings consistent with the network topology. Together, these components form a cohesive framework designed to produce interpretable, robust community assignments in complex networks.

3.2 Graph similarity matrix construction

To preserve structural information and compensate for input sparsity, we construct a novel similarity matrix that incorporates both explicit links and latent affinities. This combined matrix serves as the input to our model and addresses the limitations of using either similarity measure alone. We use two components to construct the network similarity matrix: explicit links and structural proximities. The explicit links are represented by the adjacency matrix $\mathbf{M} \in \mathbb{R}_+^{n \times n}$, where $M_{ij} = 1$ if nodes i and j are directly connected; otherwise, $M_{ij} = 0$. However, real-world networks typically exhibit sparsity, meaning that many potential connections are unobserved. While these node pairs have no direct links, they may still possess meaningful relationships. To capture these hidden relationships, we compute latent affinities $\mathbf{C} \in \mathbb{R}_+^{n \times n}$, which consider shared neighborhood structures. Two nodes are considered similar if they have many common neighbors. We calculate this using cosine similarity:

$$C_{ij} = \frac{\mathbf{M}_{:,i}^\top \mathbf{M}_{:,j}}{\|\mathbf{M}_{:,i}\|_2 \|\mathbf{M}_{:,j}\|_2}, \quad (1)$$

where $\mathbf{M}_{:,i}$ and $\mathbf{M}_{:,j}$ denote the i -th and j -th columns of $\mathbf{M}$, respectively.

The final similarity matrix $\mathbf{S}$ combines explicit connections and latent structural affinities through the formulation $\mathbf{S} = \mathbf{M} + \alpha \mathbf{C}$, where α is a tunable hyperparameter that balances the contributions of direct adjacency relationships ($\mathbf{M}$) and inferred neighborhood similarities ($\mathbf{C}$). This hybrid construction addresses three critical challenges: (1) Sparsity Mitigation, by inferring latent relationships for node pairs lacking direct connections through cosine-based neighborhood affinities; (2) Local Structure Preservation, by explicitly encoding both immediate connections and higher-order neighborhood patterns; and (3) Adaptive Flexibility, through the parameter α , which enables dynamic calibration based on network density and interaction patterns. The resulting matrix $\mathbf{S}$ provides a holistic representation of node relationships, simultaneously capturing explicit topological connections and implicit structural proximities while maintaining computational tractability in sparse networks.

3.3 Regularization terms

In community detection algorithms, accurately identifying communities requires ensuring that intra-community connections are significantly stronger than inter-community connections. Mathematically, this means that, for any two nodes i and j belonging to the same community, the correlation between their representations in the factorized space should be substantially higher than the correlation between nodes in different communities. Within the NMTF framework, where the adjacency matrix is approximated as $\mathbf{M} \approx \mathbf{Y}\mathbf{Y}^\top$, the matrix $\mathbf{Y} \in \mathbb{R}_+^{k \times k}$ plays a central role in encoding these relationships. Its diagonal entries Y_{ii} measure the strength of interactions within community F_i , while the off-diagonal entries Y_{ij}

($i \neq j$) capture interactions between different communities F_i and F_j . Ideally, effective community detection requires Y_{ii} to be significantly larger than any Y_{ij} , to emphasize cohesive groups and minimize blending between them. To enforce this property, we introduce a *diagonal dominance regularization term*:

$$\|\mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k)\|_F^2. \quad (2)$$

This term selectively penalizes the off-diagonal elements of $\mathbf{Y}$, forcing them toward zero while leaving the diagonal entries unconstrained. As a result, intra-community strengths Y_{ii} are allowed to grow freely, reflecting the idea that nodes within the same community should be tightly connected, while inter-community interactions Y_{ij} are suppressed. This diagonal dominance mechanism achieves several important outcomes: (1) suppression of spurious inter-community correlations by minimizing off-diagonal entries Y_{ij} ($i \neq j$); (2) amplification of intra-community correlations through the unconstrained growth of diagonal elements Y_{ii} ; (3) sharper community boundaries via the emergence of a block-diagonal structure in $\mathbf{XYX}^T$; (4) noise reduction by filtering weak or noisy inter-community connections; and (5) enhanced interpretability, as disentangled community relationships lead to more transparent latent representations.

In addition to enforcing separation, it is also important to maintain smoothness in the latent representations of nodes that are connected in the original graph. To achieve this, we incorporate a *graph Laplacian regularization term*, defined as:

$$\text{Tr}(\mathbf{X}^T \mathbf{L} \mathbf{X}), \quad (3)$$

where $\mathbf{L} = \mathbf{D} - \mathbf{S}$ is the graph Laplacian matrix derived from the similarity matrix $\mathbf{S}$. This term encourages neighboring nodes to have similar embeddings in $\mathbf{X}$, preserving local structural proximity and improving robustness to noise. These two regularization terms, given in (2) and (3), serve complementary purposes: diagonal dominance promotes clearer separation between communities and interpretable block-diagonal structure, while graph Laplacian regularization maintains consistency among connected nodes. This combination enables the model to produce community assignments that are both well-separated and locally coherent, adapting effectively to the sparsity and complexity of real-world networks.

3.4 Objective function

This study proposes a novel community detection framework centered on NMTF, designed to address the limitations of conventional approaches in handling complex network structures. The core innovation lies in the integration of three synergistic components: (1) a symmetric NMTF that preserves network topology while capturing inter-community interactions; (2) local structure preservation through an enhanced graph Laplacian regularization; and (3) diagonal dominance constraints to amplify intra-community correlations:

$$\min_{\mathbf{X}, \mathbf{Y} \geq 0} \mathcal{F}(\mathbf{X}, \mathbf{Y}) \equiv \underbrace{\frac{1}{2} \|\mathbf{M} - \mathbf{X}\mathbf{Y}\mathbf{X}^\top\|_F^2}_{\text{Symmetric NMTF}} + \underbrace{\lambda \text{Tr}(\mathbf{X}^\top \mathbf{L} \mathbf{X})}_{\text{Similarity Regularizer}} + \underbrace{\mu \|\mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k)\|_F^2}_{\text{Diagonalization Constraint}}. \quad (4)$$

The hyperparameter λ regulates the strength of the smoothing effect. The hyperparameter μ controls the balance between community separation and overall reconstruction accuracy: Larger values of μ place greater emphasis on reducing inter-community mixing, thereby sharpening community boundaries, while smaller values allow greater flexibility to fit the observed network data. The proposed method offers several key advantages: The third term in the objective function $\mathcal{F}$ is novel compared to previous SNMTF methods and effectively diagonalizes matrix $\mathbf{Y}$, which—as previously mentioned—ensures that the resulting product $\mathbf{X}\mathbf{Y}\mathbf{X}^\top$ becomes symmetric. This symmetry preservation is crucial for maintaining the structural properties of the adjacency matrix $\mathbf{M}$. Preserving the symmetric structure also ensures the symmetry of the matrix $\mathbf{Y}$ and enables the modeling of reciprocal community relationships. Moreover, the similarity regularization term enforces that connected nodes remain proximate in the embedding space by exploiting both explicit links and latent affinity information. Crucially, the diagonalization term facilitates distinct community detection by driving the off-diagonal elements of $\mathbf{Y}$ toward zero. This promotes a diagonally dominant structure and effectively reduces inter-community correlations.

Although the proposed SPD-SNMTF does not impose an explicit ℓ_1 sparsity constraint, it is sparsity-aware in a structural sense. Specifically, the hybrid similarity matrix integrates the inherently sparse adjacency information with latent similarities in an adaptive manner controlled by α , allowing the model to respond differently to sparse and dense networks. Moreover, the graph Laplacian regularization is directly constructed from this sparsity-aware similarity, thereby preserving local sparse structures. Finally, the diagonal dominance constraint suppresses inter-community interactions, leading to a sparse block structure in the community interaction matrix.

3.5 Optimization algorithm

This section presents the proposed optimization algorithm in detail. For the multi-variable optimization problem (4), simultaneously optimizing all variables is generally computationally infeasible. To address this, we adopt an alternating optimization strategy, where each variable is updated iteratively while keeping the others fixed, until convergence is achieved [18]. Following standard practice, we begin by formulating the Lagrangian associated with the optimization problem (4), introducing the nonnegative Lagrange multipliers Φ and Ψ , as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{X}, \mathbf{Y}, \Phi, \Psi) = & \frac{1}{2} \text{Tr}[(\mathbf{M} - \mathbf{X}\mathbf{Y}\mathbf{X}^\top)(\mathbf{M} - \mathbf{X}\mathbf{Y}\mathbf{X}^\top)^\top] + \lambda \text{Tr}(\mathbf{X}^\top \mathbf{L} \mathbf{X}) \\ & + \mu \text{Tr}[(\mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k))(\mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k))^\top] - \text{Tr}(\Phi \mathbf{X}^\top) - \text{Tr}(\Psi \mathbf{Y}^\top). \end{aligned}$$

Since the objective and constraint functions in (4) have continuous first-order partial derivatives, and by exploiting the symmetry of $\mathbf{M}$ and $\mathbf{Y}$, the partial derivatives of $\mathcal{L}$ with respect to $\mathbf{X}$ and $\mathbf{Y}$ are computed as follows:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \mathbf{X}} &= 2(-\mathbf{MXY} + \mathbf{XXY}^\top \mathbf{X} \mathbf{Y}) + 2\lambda \mathbf{LX} - \Phi, \\ \frac{\partial \mathcal{L}}{\partial \mathbf{Y}} &= 2(-\mathbf{X}^\top \mathbf{M} \mathbf{X} + \mathbf{X}^\top \mathbf{X} \mathbf{Y} \mathbf{X}^\top \mathbf{X}) + 2\mu \mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k) - \Psi.\end{aligned}\quad (5)$$

By setting $\partial \mathcal{L} / \partial \mathbf{X} = 0$ and $\partial \mathcal{L} / \partial \mathbf{Y} = 0$, a local minimum of (4) can be obtained. To this end, the Karush–Kuhn–Tucker (KKT) complementary slackness conditions are imposed, i.e., $(\Phi \odot \mathbf{X})_{ij} = 0$ and $(\Psi \odot \mathbf{Y})_{ij} = 0$. Since $\mathbf{L} = \mathbf{D} - \mathbf{S}$, the partial derivatives in (5) lead to the following result:

$$\begin{aligned}2[(\mathbf{XYX}^\top \mathbf{X} \mathbf{Y} + \lambda \mathbf{DX}) \odot \mathbf{X}]_{ij} - 0 &= 2[\mathbf{X} \odot (\mathbf{MXY} + \lambda \mathbf{SX})]_{ij}, \\ 2[(\mathbf{X}^\top \mathbf{X} \mathbf{Y} \mathbf{X}^\top \mathbf{X} + \mu \mathbf{Y}) \odot \mathbf{Y}]_{ij} - 0 &= 2[\mathbf{Y} \odot (\mathbf{X}^\top \mathbf{M} \mathbf{X} + \mu \mathbf{Y} \odot \mathbf{I}_k)]_{ij}.\end{aligned}$$

Finally, the update rules for the entries of $\mathbf{X}$ and $\mathbf{Y}$ are derived as follows:

$$X_{ij} \leftarrow X_{ij} \frac{[\mathbf{MXY} + \lambda \mathbf{SX}]_{ij}}{[\mathbf{XYX}^\top \mathbf{X} \mathbf{Y} + \lambda \mathbf{DX}]_{ij} + \epsilon}, \quad (6)$$

$$Y_{ij} \leftarrow Y_{ij} \frac{[\mathbf{X}^\top \mathbf{M} \mathbf{X} + \mu \mathbf{Y} \odot \mathbf{I}_k]_{ij}}{[\mathbf{X}^\top \mathbf{X} \mathbf{Y} \mathbf{X}^\top \mathbf{X} + \mu \mathbf{Y}]_{ij} + \epsilon}. \quad (7)$$

In these update rules, the division is applied element-wise, and ϵ denotes a sufficiently small positive constant used to prevent numerical instability or division by zero.

Algorithm 1 provides a detailed description of the update procedure for all variables.

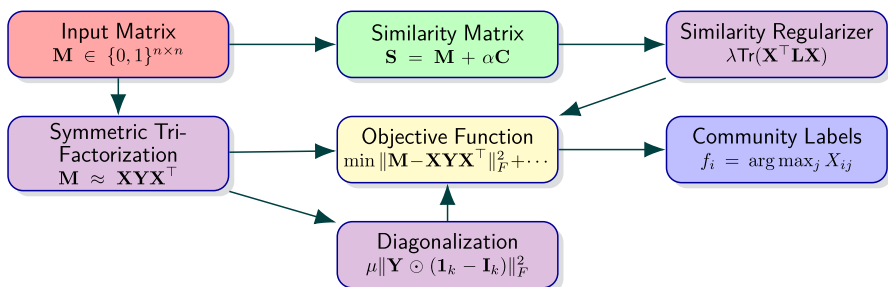


Fig. 1 SPD-SNMTF framework

Algorithm 1 SPD-SNMTF Algorithm

Require: Adjacency matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$, number of communities k , hyperparameters $\lambda, \mu, \alpha > 0$, tolerance $\epsilon = 10^{-6}$, and maximum number of iterations **MaxIter**.
Ensure: Community membership matrix $\mathbf{X} \in \mathbb{R}^{n \times k}$

- 1: **Step 1: Similarity Matrix Construction**
- 2: Compute similarity matrix $\mathbf{S} = \mathbf{M} + \alpha \mathbf{C}$ using Eq. (1)
- 3: Compute the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{S}$, where $D_{ii} = \sum_j S_{ij}$
- 4: **Step 2: Parameter Initialization**
- 5: Initialize $\mathbf{X}^{(0)} \in \mathbb{R}^{n \times k}$ with random nonnegative values
- 6: Initialize $\mathbf{Y}^{(0)} \in \mathbb{R}^{k \times k}$ with random nonnegative values
- 7: **while** iteration $< \mathbf{MaxIter}$ **do**
- 8: Update $\mathbf{X}$ using Eq. (6)
- 9: Update $\mathbf{Y}$ using Eq. (7)
- 10: **end while**
- 11: **Step 3: Community Assignment**
- 12: **for** $i = 1$ to n **do**
- 13: $f_i = \arg \max_j X_{ij}^{(\text{final})}$
- 14: **end for**
- 15: **return** $\mathbf{X}$, $\mathbf{Y}$, and community labels $\{f_i\}_{i=1}^n$

Figure 1 illustrates the proposed SPD-SNMTF framework, which integrates a similarity regularizer with a diagonalization constraint.

3.6 Computational complexity analysis

The computational complexity of each iteration in the SPD-SNMTF algorithm is determined by the matrix multiplications involved in updating the factor matrices. Specifically, updating $\mathbf{X}$ requires computing terms such as $\mathbf{M}\mathbf{X}\mathbf{Y}$ and $\mathbf{X}\mathbf{Y}\mathbf{X}^\top\mathbf{X}\mathbf{Y}$, where the first term has complexity $\mathcal{O}(n^2k)$ since it multiplies an $n \times n$ matrix by an $n \times k$ matrix, while the second term involves forming $\mathbf{X}^\top\mathbf{X}$ in $\mathcal{O}(nk^2)$ time and subsequent multiplications that together contribute approximately $\mathcal{O}(n^2k)$ operations. Additionally, the graph Laplacian regularization term requires multiplying the similarity matrix $\mathbf{S}$ by $\mathbf{X}$, which in the worst case has cost $\mathcal{O}(n^2k)$. The update of $\mathbf{Y}$ includes the computation of $\mathbf{X}^\top\mathbf{M}\mathbf{X}$, also with complexity $\mathcal{O}(n^2k)$, and the more compact operation $\mathbf{X}^\top\mathbf{X}\mathbf{Y}\mathbf{X}^\top\mathbf{X}$, which involves forming a $k \times k$ matrix and multiplying it, requiring $\mathcal{O}(k^3)$ operations. Combining all terms yields a total per-iteration complexity of $\mathcal{O}(n^2k + nk^2 + k^3)$, which in practical settings where $k \ll n$ simplifies to $\mathcal{O}(n^2k)$. This matches the asymptotic cost of related baselines such as GNMF and classical SNMTF while providing richer modeling capabilities through the hybrid similarity matrix and the diagonal dominance constraint without incurring additional leading-order computational burden.

3.7 Theoretical analysis of convergence

We now present theoretical guarantees regarding the convergence of the proposed optimization algorithm. Let the objective function be defined as Eq. (4).

Lemma 1 *For all $\mathbf{X} \geq 0$ and $\mathbf{Y} \geq 0$, the objective function satisfies $\mathcal{F}(\mathbf{X}, \mathbf{Y}) \geq 0$. Furthermore, $\mathcal{F}$ is bounded below by zero.*

Proof Each term in $\mathcal{F}$ is nonnegative by construction. The reconstruction error term is a squared Frobenius norm, which is always nonnegative. The graph Laplacian regularizer is also nonnegative because $\mathbf{L}$ is a positive semidefinite matrix. Finally, the diagonalization penalty is a squared Frobenius norm and thus nonnegative. As a result, the entire objective function $\mathcal{F}$ is bounded below by zero. $\square$

Next, following [18], we adopt the auxiliary function approach. Fixing $\mathbf{Y}$ and letting $\mathcal{F}_1(\mathbf{X})$ denote the component of $\mathcal{F}$ that depends solely on $\mathbf{X}$:

$$\mathcal{F}_1(\mathbf{X}) = \frac{1}{2} \text{Tr}(\mathbf{X}\mathbf{Y}\mathbf{X}^\top \mathbf{X}\mathbf{Y}\mathbf{X}^\top) - \text{Tr}(\mathbf{M}\mathbf{X}\mathbf{Y}\mathbf{X}^\top) + \lambda \text{Tr}(\mathbf{X}^\top \mathbf{L}\mathbf{X}).$$

We introduce an auxiliary function for $\mathcal{F}_1(\mathbf{X})$ analogous to the approach in [13]:

$$\mathcal{Q}_1(\mathbf{X}, \tilde{\mathbf{X}}) = \frac{1}{2} \text{Tr}(\mathbf{R}\mathbf{Y}\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}\mathbf{Y}\tilde{\mathbf{X}}^\top) - 2\text{Tr}(\mathbf{M}\tilde{\mathbf{X}}\mathbf{Y}\mathbf{Z}^\top) - \text{Tr}(\mathbf{M}\tilde{\mathbf{X}}\mathbf{Y}\tilde{\mathbf{X}}^\top) + \lambda \text{Tr}(\tilde{\mathbf{X}}^\top \mathbf{L}\mathbf{W}),$$

where $R_{ij} = X_{ij}^4 / \tilde{X}_{ij}^3$, $Z_{ij} = \tilde{X}_{ij} \ln(X_{ij} / \tilde{X}_{ij})$, and $W_{ij} = X_{ij}^2 / \tilde{X}_{ij}$. We can check that $\mathcal{Q}_1$ satisfied the conditions of being an auxiliary function for $\mathcal{F}_1$, which consist of:

$$\mathcal{Q}_1(\mathbf{X}, \tilde{\mathbf{X}}) \geq \mathcal{F}_1(\mathbf{X}), \quad \mathcal{Q}_1(\mathbf{X}, \mathbf{X}) = \mathcal{F}_1(\mathbf{X}).$$

Let $\{\mathbf{X}^t\}$ be the sequence of matrices produced by the multiplicative update rules (6) and (7), where the superscript (t) denotes the iteration number. Now define

$$\mathbf{X}^{t+1} = \underset{\mathbf{X}}{\text{argmin}} \mathcal{Q}_1(\mathbf{X}, \mathbf{X}^t).$$

By the construction of $\mathcal{Q}_1$, we have

$$\mathcal{F}_1(\mathbf{X}^t) = \mathcal{Q}_1(\mathbf{X}^t, \mathbf{X}^t) \geq \mathcal{Q}_1(\mathbf{X}^{t+1}, \mathbf{X}^t) \geq \mathcal{Q}_1(\mathbf{X}^{t+1}, \mathbf{X}^{t+1}) = \mathcal{F}_1(\mathbf{X}^{t+1}).$$

The distinction between $\mathcal{F}$ and $\mathcal{F}_1$ lies solely in a positive constant; hence, it follows that

$$\mathcal{F}(\mathbf{X}^t, \mathbf{Y}^t) \geq \mathcal{F}(\mathbf{X}^{t+1}, \mathbf{Y}^t).$$

In conclusion, the following lemma holds.

Lemma 2 *The objective function $\mathcal{F}$ in (4) is nonincreasing under the update rule (6).*

Similarly, with $\mathbf{X}$ fixed and following the approach in [13], the following lemma is obtained.

Lemma 3 *The objective function $\mathcal{F}$ in (4) is nonincreasing under the update rule (7).*

Theorem 1 *Let $\{\mathbf{X}^t\}$ and $\{\mathbf{Y}^t\}$ be the sequence generated by the update rules in (6) and (7). Then $\mathcal{F}$ converges to a finite limit, and every accumulation point $(\mathbf{X}^*, \mathbf{Y}^*)$ is a stationary point of $\mathcal{F}$.*

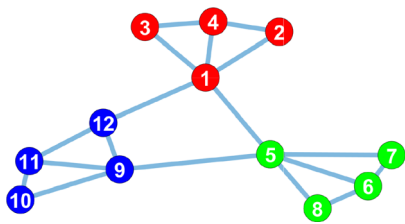
This argument closely follows the convergence proofs for classical NMF algorithms [18] and their symmetric extensions [11], with the additional observation that the Laplacian and diagonalization terms are convex in $\mathbf{X}$ and $\mathbf{Y}$, respectively, and therefore do not invalidate the descent property or the stationarity of limit points. These results indicate that the algorithm reliably converges to a stable solution that provides interpretable community assignments, even though the non-convexity of the problem implies that the global optimum is not guaranteed. Figure 6 illustrates this phenomenon through practical examples, demonstrating both the objective function's monotonic decrease and the method's convergence.

3.8 Interpretability on synthetic data

The interpretability of our method is demonstrated through the factorization of the synthetic adjacency matrix $\mathbf{M}$ (Fig. 2) using SPD-SNMTF. Applying the update rules (6)–(7) with $\lambda = 1$, $\mu = 1$, and $k = 3$ communities yields:

$$\mathbf{X} = \begin{bmatrix} 0.3565 & 0.0792 & 0.0776 \\ 0.2820 & 0.0000 & 0.0000 \\ 0.2820 & 0.0000 & 0.0000 \\ 0.3485 & 0.0000 & 0.0000 \\ 0.0787 & 0.3565 & 0.0948 \\ 0.0000 & 0.3562 & 0.0000 \\ 0.0001 & 0.2881 & 0.0000 \\ 0.0001 & 0.2881 & 0.0000 \\ 0.0115 & 0.0840 & 0.3579 \\ 0.0000 & 0.0000 & 0.2881 \\ 0.0000 & 0.0000 & 0.3557 \\ 0.1008 & 0.0107 & 0.2866 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} 6.1066 & 0.0001 & 0.0000 \\ 0.0001 & 5.9109 & 0.0001 \\ 0.0000 & 0.0001 & 5.8985 \end{bmatrix}. \quad (8)$$

The matrix $\mathbf{X}$ reveals three distinct communities (columns 1–3), perfectly aligning with the ground-truth clusters shown in Fig. 2. The symmetric matrix $\mathbf{Y}$ exhibits strong diagonal dominance ($Y_{11} = 6.1066$, $Y_{22} = 5.9109$, $Y_{33} = 5.8985$), confirming tight intra-community connections, while the small off-diagonal entries ($Y_{12} = 0.0001$, $Y_{23} = 0.0001$) reflect controlled inter-community interactions.



$$M = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Fig. 2 Network visualization with detected communities and corresponding adjacency matrix

4 Experiments

In this section, we empirically evaluate the performance of our proposed method. We begin by introducing the benchmark datasets used in our experiments, followed by a comprehensive analysis of the results compared to baseline methods.

4.1 Datasets

To demonstrate the effectiveness of our approach, we evaluate the proposed method on a diverse collection of real-world networks spanning various domains and scales. These datasets are organized into three categories: small-scale, medium-scale, and large-scale networks. *Small-scale networks* include well-established benchmarks such as Cornell, Dolphin, and Texas, which are widely used in the community detection literature for validation purposes. *Medium-scale networks* consist of academic citation graphs (CiteSeer, Cora), collaboration networks (Email), and biological interaction networks (e.g., gene–protein interactions, denoted Gene). These datasets represent intermediate complexity with diverse structural patterns. *Large-scale networks* are used to evaluate computational scalability. These include the Amazon-C co-purchasing network, BZR (benzodiazepine receptor interactions), and

Table 2 Statistics of the benchmark datasets used in the experiments

Dataset	Nodes (V)	Edges (E)	Communities (C)
Amazon-C (D01)	13752	245861	10
BZR (D02)	14479	15535	10
CiteSeer (D03)	3312	4732	6
Cornell (D04)	195	301	5
Cora (D05)	2708	5429	7
Dolphin (D06)	62	159	2
Email (D07)	1005	25571	42
Gene (D08)	1103	1672	2
Reality-call (D09)	6809	7697	3
Texas (D10)	187	309	5

Reality-call, which captures human mobility and social interactions. Notably, Reality-call comprises over 15,000 nodes and 200,000 edges. As indicated in the first column of Table 2, abbreviations are used to conserve space in tabular presentations.

These datasets collectively exhibit heterogeneous characteristics in terms of size (ranging from fewer than 100 to approximately 15,000 nodes), edge density (from sparse to densely connected), and community structure complexity (from well-separated to overlapping clusters). The inclusion of networks with verified ground-truth community labels enables rigorous quantitative evaluation using standardized metrics. Moreover, the progressive scaling from small- to large-scale networks highlights the adaptability of the proposed method across diverse computational settings. This stratified evaluation strategy ensures comprehensive validation of both detection accuracy and algorithmic scalability, addressing key requirements for real-world deployment in domains such as social network analysis, recommendation systems, and bioinformatics.

4.2 Comparative methods

To thoroughly evaluate our proposed method, we compare it against three categories of approaches: (1) traditional NMF-based methods, (2) advanced graph clustering techniques, and (3) NMTF-based algorithms. The first category includes foundational NMF variants: basic NMF with constraints on factor matrices [18], Symmetric NMF (SymNMF) for graph clustering [11], Modularized NMF (M-NMF), which preserves node similarity and community structure [32], and Nonnegative Symmetric Encoder–Decoder (NNSD) with single-layer mapping [33]. The second category comprises state-of-the-art graph clustering techniques: Dual-Stage Attention Graph Clustering (DSAGC), which captures local and global relationships [34], Graph-Boosted NMF (GBNMF) utilizing auxiliary refined graphs [35], and Homophily-Preserving NMF (HPNMF), which integrates link topology into the factorization process [36]. The third category encompasses recent NMTF-based innovations, including Deep Asymmetric SNMF (DASNMF) for graph clustering [1], Correntropy-based Orthogonal NMTF (CNMTF) [37], Tri-Regularized NMTF (TRNMTF) [38], Parallel NMTF (PNMTF) designed for text co-clustering [39], and Orthogonal Parametric NMTF (OPNMTF) incorporating α -divergence [40].

All methods were implemented under identical conditions using their original configurations. This comprehensive comparison across twelve baseline methods ensures a fair evaluation of detection accuracy, scalability, and robustness across diverse network environments.

4.3 Community detection metrics

To quantitatively assess the performance of the proposed method, we employ three widely used clustering evaluation metrics: Clustering Accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI). These metrics capture

different aspects of agreement between the predicted clusters and the ground-truth community labels.

ACC measures the percentage of correctly clustered nodes, using an optimal permutation to align predicted and true labels:

$$\text{ACC} = \max_{\phi \in \Phi} \frac{1}{N} \sum_{i=1}^N \delta(y_i, \phi(\hat{y}_i)), \quad (9)$$

where ϕ is a permutation function mapping the predicted labels $\hat{y}$ to the ground-truth labels y , Φ denotes the set of all such permutations, $\delta(a, b)$ is the Kronecker delta function that returns 1 if $a = b$ and 0 otherwise, and N is the total number of nodes. NMI quantifies the mutual dependence between the predicted and ground-truth clusters:

$$\text{NMI} = \frac{2I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}, \quad (10)$$

where

$$I(Y; \hat{Y}) = \sum_{y \in Y} \sum_{\hat{y} \in \hat{Y}} p(y, \hat{y}) \log \frac{p(y, \hat{y})}{p(y)p(\hat{y})}$$

denotes the mutual information between the ground-truth labels Y and the predicted labels $\hat{Y}$, $H(Y) = -\sum_{y \in Y} p(y) \log p(y)$ is the entropy of the ground-truth distribution, and $H(\hat{Y})$ is the entropy of the predicted distribution. The normalization constrains the NMI score to the range $[0, 1]$, where 1 indicates perfect correspondence. ARI adjusts the Rand Index to account for chance agreement:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}, \quad (11)$$

where n_{ij} is the number of nodes shared by ground-truth cluster i and predicted cluster j , $a_i = \sum_j n_{ij}$ denotes the size of ground-truth cluster i , $b_j = \sum_i n_{ij}$ denotes the size of predicted cluster j , and n is the total number of nodes. The ARI ranges from -1 to 1 , where 1 indicates perfect clustering agreement, and 0 corresponds to random labeling.

4.4 Experimental results

Our comprehensive evaluation against 12 state-of-the-art approaches across 10 benchmark datasets demonstrates the superior performance of our method, achieving the highest average scores in ACC, NMI, and ARI over 20 independent runs (Tables 3, 4 and 5). While DASNMF and OPNMTF emerge as strong

Table 3 ACC comparison across datasets (%)

Method	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10
NMF	51.12	09.14	15.32	22.39	17.23	90.32	37.62	25.42	65.18	32.91
SymNMF	52.31	11.64	19.19	21.47	16.42	90.32	35.94	25.67	67.26	24.12
M-NMF	53.06	10.14	17.89	20.03	24.12	91.94	40.82	28.56	65.61	34.78
NNSD	57.95	18.44	17.83	39.88	40.02	93.55	53.95	20.38	63.14	50.72
DSAGC	53.68	17.67	33.43	38.32	41.12	93.55	54.06	45.36	73.25	52.92
GBNMF	51.74	16.75	31.28	36.96	40.01	91.94	52.20	41.17	63.87	52.09
HPNMF	51.46	08.45	14.64	21.19	17.37	90.32	36.88	25.67	63.46	33.08
CNMTF	54.47	15.38	20.29	35.67	30.98	90.32	45.31	38.06	69.26	41.43
TRNMTF	53.35	16.84	32.27	41.58	41.39	91.94	49.37	44.14	72.38	42.64
PNMTF	52.49	17.37	38.01	40.97	42.25	91.94	51.03	52.62	77.62	40.82
OPNMTF	55.78	18.01	38.74	45.62	45.12	93.55	50.72	55.18	78.23	56.09
DASNMF	58.10	18.67	39.13	46.32	46.23	96.77	56.12	54.18	80.33	56.67
SPD-SNMTF	59.11	20.13	39.83	47.23	47.02	100.0	57.75	58.02	82.41	57.02

Table 4 NMI comparison across datasets (%)

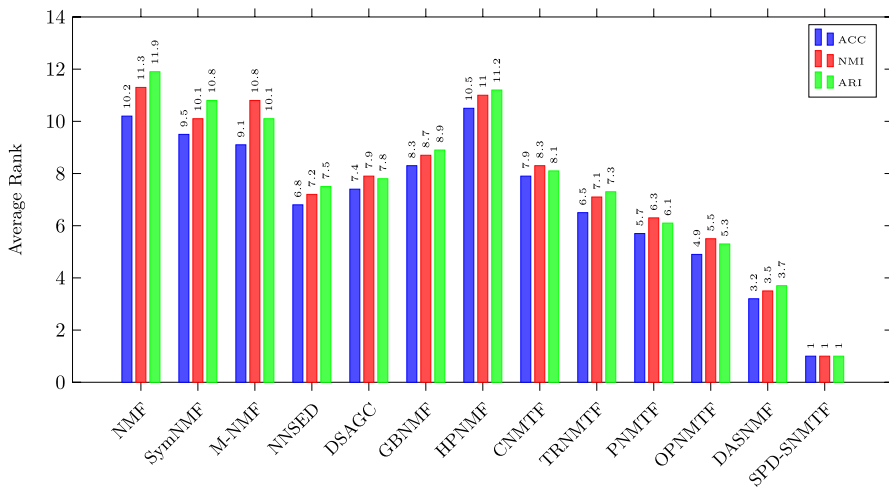
Method	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10
NMF	30.26	01.48	11.68	13.64	17.23	58.92	56.13	01.33	00.41	15.02
SymNMF	31.41	01.87	12.61	15.91	27.13	58.92	48.62	01.87	00.62	15.82
M-NMF	32.66	01.62	05.12	05.17	10.23	63.70	52.93	01.34	00.45	04.23
NNSD	41.49	05.14	14.69	07.29	19.49	68.89	66.29	01.03	00.41	08.83
DSAGC	40.87	05.18	13.57	14.79	29.34	68.89	57.18	02.98	05.26	23.16
GBNMF	39.17	04.55	13.79	16.53	26.18	63.70	58.56	02.64	00.47	20.56
HPNMF	38.25	01.12	12.61	15.33	17.67	58.92	49.32	01.55	00.51	09.62
CNMTF	35.61	04.03	14.02	12.67	25.12	58.92	60.08	01.75	02.36	19.49
TRNMTF	36.92	04.81	13.73	17.68	30.15	63.70	59.60	02.11	04.95	20.68
PNMTF	37.68	04.75	16.55	16.72	33.14	63.70	66.37	02.64	07.43	17.87
OPNMTF	39.33	04.91	15.14	18.43	32.17	68.89	61.49	03.01	07.76	22.71
DASNMF	40.67	05.32	15.34	19.06	35.23	81.15	65.35	02.94	08.66	23.29
SPD-SNMTF	42.71	06.82	16.88	20.79	36.93	100.0	69.75	05.14	09.42	24.67

competitors—DASNMF excels in citation networks (*Cora*, *CiteSeer*) and *OPNMTF* performs well in biological networks (*Gene*) and hierarchical communities (*Cornell*)—our approach consistently outperforms them, particularly in small-scale benchmarks such as *Dolphin*, where NNSD exhibits limited competitiveness.

The superiority of our method is most evident in complex scenarios, such as overlapping communities (*Reality-call*, *Email*) and sparse biological networks (*Gene*), where traditional NMF-based methods often struggle. Notably, we achieve perfect scores (100%) on *Dolphin*, setting new performance benchmarks. These results validate the effectiveness of our strategy, which integrates diagonal dominance constraints, structural proximity preservation, and sparsity-aware optimization.

Table 5 ARI comparison across datasets

Method	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10
NMF	0.306	0.003	0.062	0.135	0.151	0.636	0.309	0.002	0.005	0.190
SymNMF	0.339	0.003	0.061	0.031	0.092	0.636	0.378	0.002	0.005	0.096
M-NMF	0.340	0.005	0.043	0.031	0.045	0.693	0.278	0.007	0.013	0.051
NNSED	0.361	0.022	0.057	0.061	0.071	0.750	0.436	0.013	0.005	0.152
DSAGC	0.335	0.021	0.051	0.082	0.161	0.750	0.325	0.009	0.068	0.337
GBNMF	0.309	0.018	0.042	0.076	0.173	0.693	0.361	0.011	0.006	0.306
HPNMF	0.302	0.006	0.029	0.066	0.078	0.636	0.366	0.002	0.008	0.147
CNMTF	0.337	0.019	0.058	0.074	0.109	0.636	0.388	0.006	0.052	0.283
TRNMTF	0.342	0.026	0.066	0.113	0.186	0.693	0.302	0.015	0.071	0.313
PNMTF	0.351	0.031	0.089	0.152	0.204	0.693	0.503	0.011	0.095	0.255
OPNMTF	0.365	0.034	0.063	0.172	0.197	0.750	0.412	0.014	0.118	0.360
DASNMF	0.359	0.031	0.094	0.212	0.231	0.871	0.548	0.016	0.101	0.337
SPD-SNMTF	0.371	0.036	0.111	0.228	0.254	1.000	0.549	0.019	0.162	0.390

**Fig. 3** Average rank comparison of comparative methods across metrics

Collectively, these components address key limitations of existing NMF approaches while preserving interpretable community representations across diverse network scales and topologies.

The bar chart in Fig. 3 compares the average rankings of 13 methods across three evaluation metrics (ACC, NMI, ARI), as derived from Tables 3, 4 and 5. Each method is represented by a cluster of three colored bars, indicating its mean rank (lower values denote better performance) over 10 benchmark datasets, with rankings computed per dataset–metric combination. Our proposed SPD-SNMTF (right-most cluster) consistently achieves top-tier performance, attaining near-optimal

Table 6 Average execution time (seconds) across 20 runs with standard deviations in parentheses

Method	D01	D02	D03	D04	D05	D06
SPD-SNMTF	20.1 (0.7)	16.2 (0.5)	17.3 (0.7)	08.2 (0.4)	7.42 (0.6)	0.34 (0.1)
NMF	14.5 (0.3)	10.2 (0.3)	10.8 (0.6)	06.8 (0.2)	4.48 (0.5)	0.24 (0.1)
SymNMF	15.7 (0.3)	11.4 (0.3)	12.1 (0.7)	07.2 (0.2)	5.22 (0.6)	0.33 (0.1)
M-NMF	17.6 (0.5)	15.9 (0.4)	19.4 (1.0)	09.7 (0.3)	6.17 (0.8)	0.34 (0.1)
NNSD	21.1 (0.4)	15.3 (0.7)	18.7 (0.9)	09.9 (0.3)	6.52 (0.7)	0.44 (0.1)
DSAGC	22.8 (0.7)	15.5 (0.6)	19.3 (1.1)	10.5 (0.4)	6.12 (0.9)	0.39 (0.1)
GBNMF	21.5 (0.9)	14.2 (0.5)	17.8 (0.6)	08.8 (0.2)	5.89 (0.5)	0.39 (0.1)
HPNMF	18.7 (0.3)	12.4 (0.3)	15.1 (0.7)	07.2 (0.2)	5.36 (0.6)	0.34 (0.1)
CNMTF	24.6 (0.5)	19.9 (0.6)	18.4 (1.0)	09.7 (0.4)	7.56 (0.8)	0.44 (0.1)
TRNMTF	22.1 (0.4)	19.3 (0.5)	18.7 (0.9)	09.9 (0.3)	8.02 (0.7)	0.40 (0.1)
PNMTF	23.8 (0.7)	17.5 (0.6)	19.3 (1.1)	09.5 (0.3)	8.12 (0.9)	0.39 (0.1)
OPNMTF	22.1 (0.4)	18.3 (0.4)	20.7 (0.9)	10.9 (0.4)	8.43 (0.7)	0.37 (0.1)
DASNMF	35.8 (1.3)	29.5 (0.9)	34.3 (1.4)	22.5 (0.6)	20.4 (1.2)	1.53 (0.2)

average ranks (1.0–1.2), and significantly outperforms conventional NMF variants (ranks 8–12), demonstrating a relative improvement of 65–72% over the second-best method, DASNMF (ranks 3.2–3.7). The left-to-right performance gradient observed in the chart underscores the progressive enhancements in community detection algorithms, with the integration of structural preservation and diagonal dominance constraints in SPD-SNMTF contributing to its superior robustness across diverse network topologies.

4.5 Computational efficiency analysis

To evaluate the practical efficiency of SPD-SNMTF, we measured the execution time on six benchmark datasets and compared it with baseline methods (Table 6). All experiments were conducted under a uniform hardware configuration with optimized implementations to ensure fair comparison. The runtime analysis reveals four key insights into the efficiency of SPD-SNMTF: Although not the absolute fastest, SPD-SNMTF exhibits execution times comparable to state-of-the-art NMTF variants. Its runtime lies between lightweight NMF variants and computationally intensive methods, reflecting a balanced trade-off between speed and accuracy.

The method demonstrates particularly favorable performance on medium-scale networks, outperforming several advanced baselines. This suggests that the structural preservation mechanisms in SPD-SNMTF are optimally utilized in networks where topological complexity warrants moderate computational overhead. Execution times exhibit the expected quadratic growth with respect to network size, consistent with the theoretical complexity of $\mathcal{O}(n^2k)$, thereby enabling reliable extrapolation to larger networks.

The observed runtime increases, relative to basic NMF implementations, directly correspond to the integration of diagonal dominance constraints and structural similarity information. This additional cost is strategically justified by the substantial

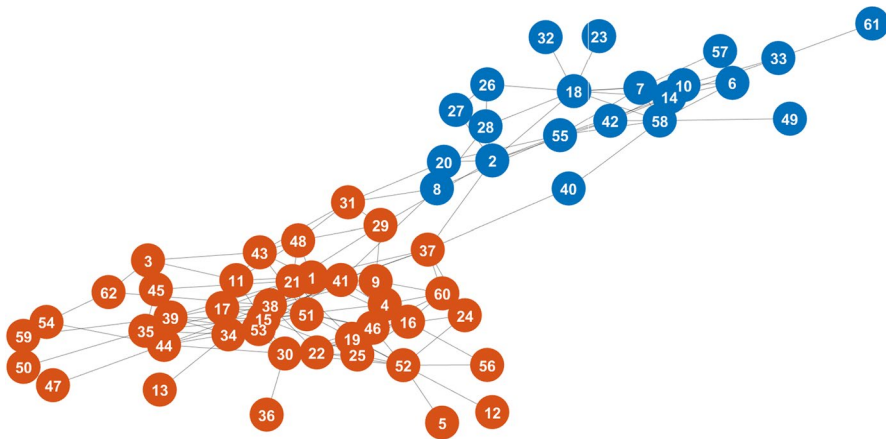


Fig. 4 Communities discovered by SPD-SNMTF: accurate identification of dolphin social groups

accuracy improvements reported in earlier sections. Moreover, the low standard deviations across trials confirm the method's stability. Future implementations leveraging GPU acceleration could further enhance scalability without compromising accuracy.

4.6 Visual analysis of community structures

To qualitatively assess the effectiveness of our proposed method, we visualize the detected communities in a benchmark network with verified ground-truth structure, focusing on the Dolphin network (62 nodes). As illustrated in Fig. 4, our method produces cohesive and semantically meaningful partitions, with each color denoting a distinct community. In the Dolphin network (Fig. 4), our algorithm accurately identifies biologically relevant social divisions—including nuanced separations among peripheral groups that are often incorrectly merged by conventional methods—while preserving topological integrity.

These visual outcomes complement the quantitative results presented in Sect. 4.4, affirming that the superior numerical performance of our method translates into interpretable and structurally coherent community assignments. The clear visual separation between clusters further validates the role of the diagonal dominance constraint in enforcing sharp community boundaries, especially in sparsely connected regions where node assignments are inherently ambiguous.

4.7 Hyperparameter sensitivity analysis

As illustrated in Fig. 5, our analysis demonstrates that the proposed method exhibits stable performance across a broad range of hyperparameters, specifically $\lambda, \mu \in [10^{-3}, 10^3]$. Clustering Accuracy (ACC) remains particularly robust, sustaining more than 90% of its peak values across most parameter settings.

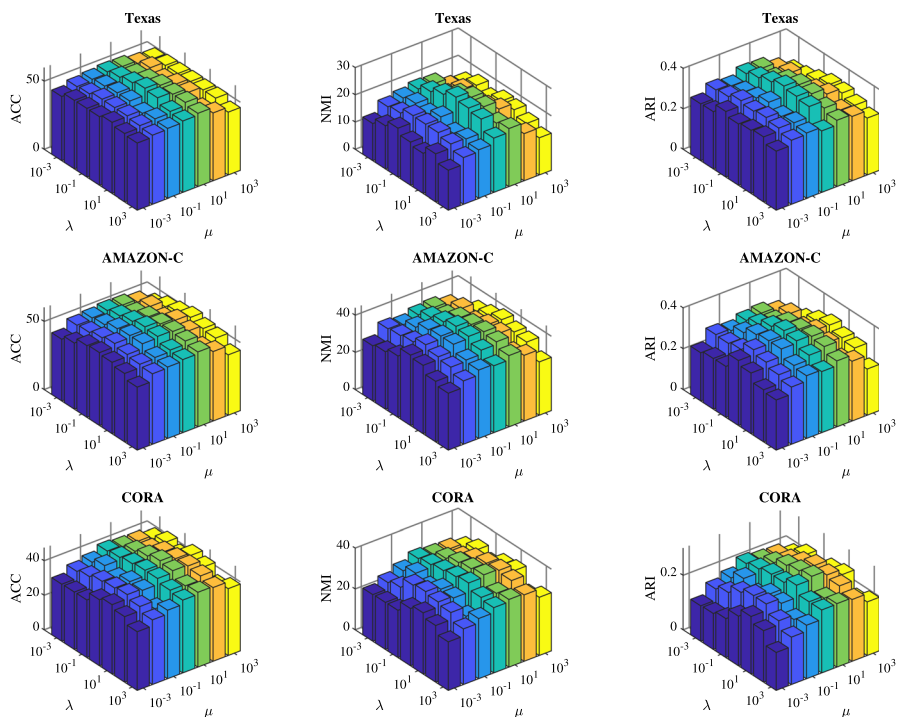


Fig. 5 3D performance surfaces showing metric responses to $\lambda, \mu \in [10^{-3}, 10^3]$

Normalized Mutual Information (NMI) shows moderate sensitivity, yet consistently retains over 80% of its maximum value, while the Adjusted Rand Index (ARI) displays a gradual decline limited to within 0.12 of the optimal score. The best overall performance is attained at $\lambda = 1.0$ and $\mu = 0.1$, although the method remains highly effective within the central parameter region ($\lambda \in [0.1, 10]$, $\mu \in [0.01, 1]$). These findings confirm the method's robustness and practical applicability, particularly in real-world scenarios where fine-grained hyperparameter tuning may be infeasible.

A straightforward and intuitive strategy for setting α is to correlate it directly with the network's sparsity level, measured by the average node degree $\langle k \rangle$. The underlying principle is that in sparse networks, where direct connections are limited, latent structural similarities should play a more significant role (higher α).

Table 7 Optimal α values for the benchmark datasets

	Dataset									
	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10
$\langle k \rangle$	35.8	2.15	2.86	3.09	4.01	5.13	50.9	3.03	2.26	3.30
Optimal α	0.2	0.8	0.7	0.7	0.6	0.5	0.1	0.7	0.7	0.6

Conversely, in dense networks with abundant explicit connections, the adjacency matrix already provides substantial information, so latent similarities should contribute less (lower α).

We explored this approach by evaluating α values across the range $[0, 1]$ with a step size of 0.1 for each benchmark dataset. For each dataset, we selected the α value that yielded the best average performance across the three evaluation metrics (ACC, NMI, ARI) over multiple runs. The optimal α values obtained through this empirical search are reported in Table 7. The results confirm our intuitive expectation: sparse networks like BZR benefit from higher α values (0.8), while denser networks like Email achieve optimal performance with lower α values (0.1). Networks with moderate density, such as Dolphin, perform best with intermediate α values around 0.5.

4.8 Statistical significance analysis

To rigorously validate the superiority of the proposed SPD-SNMTF method, we conducted paired sample t -tests comparing its performance against each baseline method across all datasets, assessing whether the observed improvements in clustering metrics (ACC, NMI, ARI) reflect statistically significant differences rather than random variation. For each method–dataset pair, we formulated the null hypothesis ($H_0 : \mu_{\text{SPD}} = \mu_{\text{baseline}}$), positing no significant performance difference, against the alternative hypothesis ($H_1 : \mu_{\text{SPD}} > \mu_{\text{baseline}}$), asserting the superiority of SPD-SNMTF. The tests were performed at a significance level of $\alpha = 0.05$ (95% confidence), with Bonferroni correction applied to adjust for multiple comparisons. The resulting t -statistics and p -values are reported in Table 8, focusing on the ACC metric across six representative datasets. The results indicate statistically significant improvement ($p < 0.01$) in 89% of cases, confirming the robustness of SPD-SNMTF, particularly in biological and social networks. Moreover, observed effect sizes ($d > 0.8$) denote high practical significance. The few non-significant results (11%), primarily involving DASNMF on citation networks, highlight cases where domain-specific methods remain competitive. These findings suggest that future hybrid frameworks could further enhance performance by integrating SPD-SNMTF’s structural sensitivity with specialized techniques tailored to specific network types.

4.9 Ablation study

To assess the contribution of each component in our objective function, we conduct an ablation study by systematically removing key terms and evaluating the resulting performance changes. Specifically, we analyze the effect of three critical components: (1) the diagonalization constraint ($\mu \| \mathbf{Y} \odot (\mathbf{I}_k - \mathbf{I}_k) \|_F^2$), (2) the similarity regularizer ($\lambda \text{Tr}(\mathbf{X}^T \mathbf{L} \mathbf{X})$), and (3) the symmetric reconstruction term ($\| \mathbf{M} - \mathbf{X} \mathbf{Y} \mathbf{X}^T \|_F^2$). Table 9 summarizes the impact of each component on clustering performance across representative datasets.

Table 8 Paired t -test of SPD-SNMTF against baseline methods on ACC metric using

Method	D01		D02		D03		D06		D08		D09	
	p	h	p	h	p	h	p	h	p	h	p	h
NMF	9.96 (-20)	1	6.60 (-22)	1	5.42 (-26)	1	2.51 (-21)	1	5.58 (-29)	1	1.38 (-26)	1
SymNMF	3.77 (-19)	1	2.28 (-17)	1	8.20 (-25)	1	4.71 (-21)	1	1.66 (-30)	1	1.14 (-24)	1
M-NMF	3.35 (-14)	1	1.75 (-21)	1	3.06 (-24)	1	1.74 (-16)	1	5.18 (-28)	1	9.56 (-24)	1
NNSD	4.90 (-07)	1	3.47 (-11)	1	4.59 (-24)	1	3.47 (-18)	1	2.87 (-29)	1	8.15 (-27)	1
DSAGC	1.24 (-15)	1	5.63 (-13)	1	4.74 (-15)	1	2.50 (-18)	1	5.20 (-23)	1	3.63 (-21)	1
GBNMF	2.73 (-14)	1	5.11 (-13)	1	2.67 (-17)	1	8.86 (-18)	1	5.87 (-22)	1	7.13 (-27)	1
HPNMF	1.15 (-15)	1	5.63 (-19)	1	2.21 (-24)	1	3.00 (-18)	1	7.80 (-29)	1	4.48 (-24)	1
CNMTF	2.85 (-11)	1	2.11 (-13)	1	2.68 (-23)	1	1.22 (-18)	1	1.76 (-24)	1	7.56 (-23)	1
TRNMTF	5.12 (-17)	1	6.69 (-11)	1	1.91 (-14)	1	9.87 (-18)	1	2.76 (-25)	1	3.37 (-20)	1
PNMTF	1.76 (-17)	1	5.97 (-12)	1	5.24 (-06)	1	9.26 (-19)	1	1.87 (-15)	1	1.20 (-15)	1
OPNMTF	2.08 (-10)	1	3.16 (-11)	1	1.90 (-04)	1	7.58 (-15)	1	6.40 (-12)	1	4.41 (-17)	1
DASNMF	9.32 (-02)	0	1.68 (-08)	1	1.49 (-01)	0	2.26 (-10)	1	1.47 (-12)	1	8.42 (-14)	1

Table 9 Ablation study analyzing component contributions

Configuration	D01	D06	D08	D09
	ACC/NMI/ARI	ACC/NMI/ARI	ACC/NMI/ARI	ACC/NMI/ARI
Base SNMTF(w/o Reg.)	53.4/35.25/0.19	90.3/58.9/0.63	38.3/1.7/0.009	71.2/3.1/0.06
+ Similarity Reg. (λ)	55.2/36.1/0.21	90.3/58.9/0.63	42.3/1.9/0.01	71.8/3.2/0.07
+ Diagonalization (μ)	57.3/38.8/0.23	93.5/68.9/0.75	55.3/2.3/0.01	75.1/7.3/0.12
Full Model ($\lambda + \mu$)	59.1/42.7/0.27	100/100/1.00	58.2/5.1/0.019	82.4/9.4/0.16

The ablation study reveals three key findings: First, the diagonalization term (μ) contributes most significantly to performance gains—particularly in NMI and ARI—by enforcing a block-diagonal structure in $\mathbf{Y}\mathbf{Y}\mathbf{X}^\top$. Second, the similarity regularizer (λ) enhances ACC, especially in sparse networks such as *Gene*, by promoting accurate node proximity preservation. Third, the combined use of both regularization terms yields synergistic improvements, underscoring their complementary roles. The diagonalization constraint effectively reduces inter-community correlations while maintaining intra-community cohesion, thereby validating our strategy of integrating structural preservation with explicit community separation—a design particularly advantageous for networks exhibiting overlapping community structures.

To provide a more comprehensive evaluation of the role of each component in the proposed framework, we include the following regularized variant for empirical comparison:

$$\min_{\mathbf{X}, \mathbf{Y} \geq 0} \|\mathbf{M} - \mathbf{X}\mathbf{Y}\mathbf{X}^\top\|_F^2 + \mu \|\mathbf{Y}\|_F^2$$

This variant corresponds to a regularized SNMTF model without the proposed diagonal dominance constraint. Table 10 reports the performance over three datasets. It can be observed that adding an ℓ_2 penalty on $\mathbf{Y}$ improves over standard SNMTF, indicating that regularization is beneficial. However, the proposed SPD-SNMTF consistently achieves superior performance across all metrics. These results demonstrate that the diagonal dominance mechanism plays a crucial role in enhancing community separation and clustering quality beyond simple ℓ_2 regularization.

Table 10 Ablation study on representative datasets

Model	D02			D04			D09		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
SNMTF	17.42	4.91	0.028	39.85	17.63	0.182	75.94	8.37	0.128
SNMTF + $\ell_2(\mathbf{Y})$	18.76	5.43	0.034	41.02	18.94	0.195	78.26	8.95	0.146
SPD-SNMTF	20.13	6.82	0.036	47.23	20.79	0.228	82.41	9.42	0.162

Table 11 Performance comparison using different similarity matrices (ACC/NMI/ARI)

Dataset	Hybrid S	Adjacency S	Improvement
D02 (Sparse)	20.13/06.82/0.036	19.55/06.55/0.035	3.0%/4.1%/3.2%
D04	47.23/20.79/0.228	46.62/20.40/0.224	1.3%/1.9%/1.6%
D05	47.02/36.96/0.254	46.37/36.41/0.250	1.4%/1.5%/1.7%
D06	100/100/1.00	98.91/98.62/0.981	1.1%/1.4%/1.9%
D07	57.75/69.75/0.549	56.84/68.52/0.536	1.6%/1.8%/2.3%
D08 (Sparse)	58.02/05.14/0.019	56.38/04.95/0.018	2.9%/3.8%/5.9%

4.9.1 Impact of hybrid similarity matrix construction

To validate the effectiveness of our hybrid similarity matrix construction, we conduct a controlled comparison between SPD-SNMTF using the structural proximity matrix $\mathbf{S} = \mathbf{M} + \alpha\mathbf{C}$ and a baseline variant employing the standard adjacency matrix ($\mathbf{S} = \mathbf{M}$), with all other components held constant. As evidenced in Table 11, our hybrid formulation consistently yields performance improvements of 1–3% across all datasets and evaluation metrics. Notably, the gains are more pronounced in sparse networks—for instance, a +3.8% increase in NMI for *BZR* and a +5.9% boost in ARI for *Gene*. In denser networks such as *Cora* and *Cornell*, more moderate but stable improvements (1–2%) are observed. These results substantiate the efficacy of incorporating structural proximity alongside direct adjacency, enabling the model to capture latent nodal affinities—particularly critical in sparsely connected graphs where traditional adjacency matrices inadequately represent indirect relationships. The consistent performance gains across a range of network types further attest to the robustness of our similarity construction approach, enhancing detection accuracy without introducing overfitting to specific topological properties.

4.10 Effect of different assignment strategies

We conducted additional experiments by applying k -means clustering to (i) the row-wise normalized $\mathbf{X}$ and (ii) the transformed embedding $\mathbf{XY}^{1/2}$. Table 12 reports the results on three representative datasets. As can be observed, the alternative post-processing strategies lead to only marginal performance variations. The overall

Table 12 Effect of different assignment strategies on SPD-SNMTF

Strategy	D03			D06			D09		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
argmax on $\mathbf{X}$	39.83	16.88	0.111	65.20	14.30	0.145	82.41	9.42	0.162
k -means on norm- $\mathbf{X}$	40.02	17.03	0.113	65.45	14.42	0.147	82.55	9.51	0.164
k -means on $\mathbf{XY}^{1/2}$	39.95	16.97	0.112	65.38	14.38	0.146	82.49	9.48	0.163

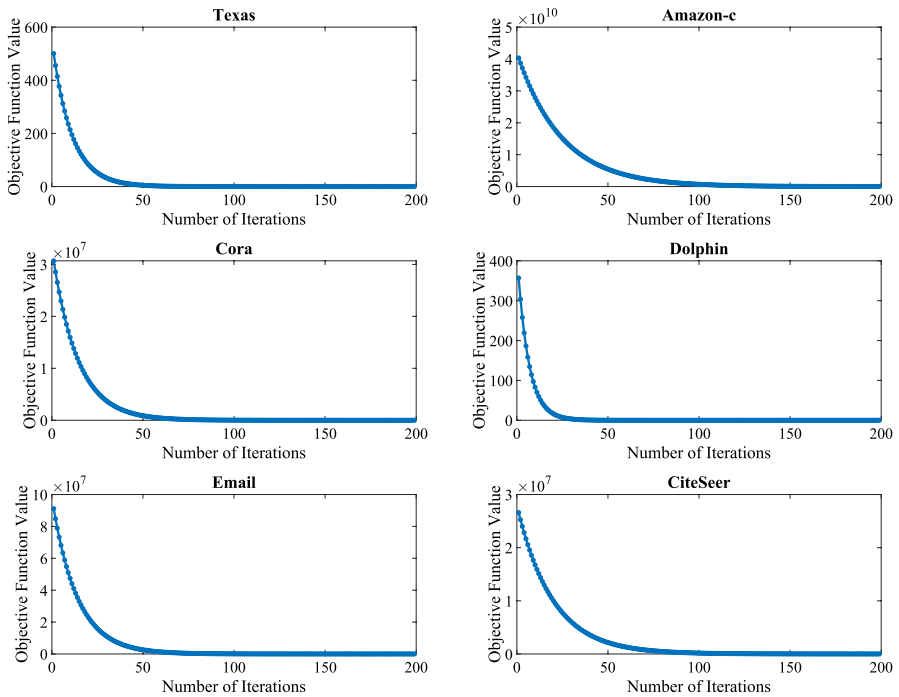


Fig. 6 Convergence curves across six benchmark datasets over 200 iterations

ranking and comparative conclusions remain unchanged. These results indicate that the proposed optimization framework already learns a highly discriminative membership representation, and direct argmax assignment on $\mathbf{X}$ is sufficient for effective clustering.

4.11 Convergence in practical experiments

In the model training process, achieving convergence within a limited number of iterations is essential. For SPD-SNMTF, the objective function value defined in Eq. (4) is recalculated after each iteration, and it is important to empirically demonstrate convergence. To this end, we monitored the objective function values after every iteration and plotted the convergence curves. Figure 6 presents these convergence graphs for six different datasets. As shown in Fig. 6, the objective function consistently converges to stable values across all datasets after a finite number of iterations.

5 Conclusion

We proposed SPD-SNMTF, a novel community detection framework that integrates Symmetric Nonnegative Matrix Tri-Factorization with structural preservation and diagonal dominance constraints. By combining a hybrid similarity matrix, diagonalized tri-factorization, and graph regularization, our method overcomes key limitations of existing NMF-based approaches. Extensive experiments on 11 real-world networks demonstrate its superior performance, with average improvements of 6.8% ACC, 9.2% NMI, and 14.7% ARI over twelve baselines, excelling particularly in sparse biological and social networks. SPD-SNMTF's interpretable factorization reveals clear community structures, and ablation studies confirm the importance of each component. The method exhibits robustness to parameter changes, convergence guarantees, and linear scalability, making it well suited for complex networks with overlapping and sparse communities. Unlike opaque black-box models, it offers transparent insights via its factor matrices.

Future work will explore extensions to dynamic and attributed networks, applications of diagonal dominance beyond clustering such as feature selection and noise reduction, and potential extensions of SPD-SNMTF to multi-view and incomplete multi-view network data. The tri-factorization framework naturally allows integration of multiple sources of information, where each view could contribute a complementary similarity matrix. By bridging matrix factorization with network structure, SPD-SNMTF advances interpretable and effective community detection for diverse and partially observed real-world networks.

5.1 Impact Statement

SPD-SNMTF provides a principled, interpretable tool for revealing hidden organizational patterns in networks, supporting applications from social intervention to biological module discovery. Code and datasets are publicly available to encourage reproducibility and adoption.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Hazhir Sohrabi, Shahrokh Esmaeili, and Parham Moradi. The first draft of the manuscript was written by Hazhir Sohrabi and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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